INCH-POUND

AN818 Rev 20
w/AMENDMENT 4
26 September 2024
SUPERSEDING
AN818 Rev 20
w/AMENDMENT 3
15 August 2019

DETAIL SPECIFICATION SHEET

NUT, TUBE COUPLING, SHORT

This specification is approved for use by all Departments and Agencies
of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4841.

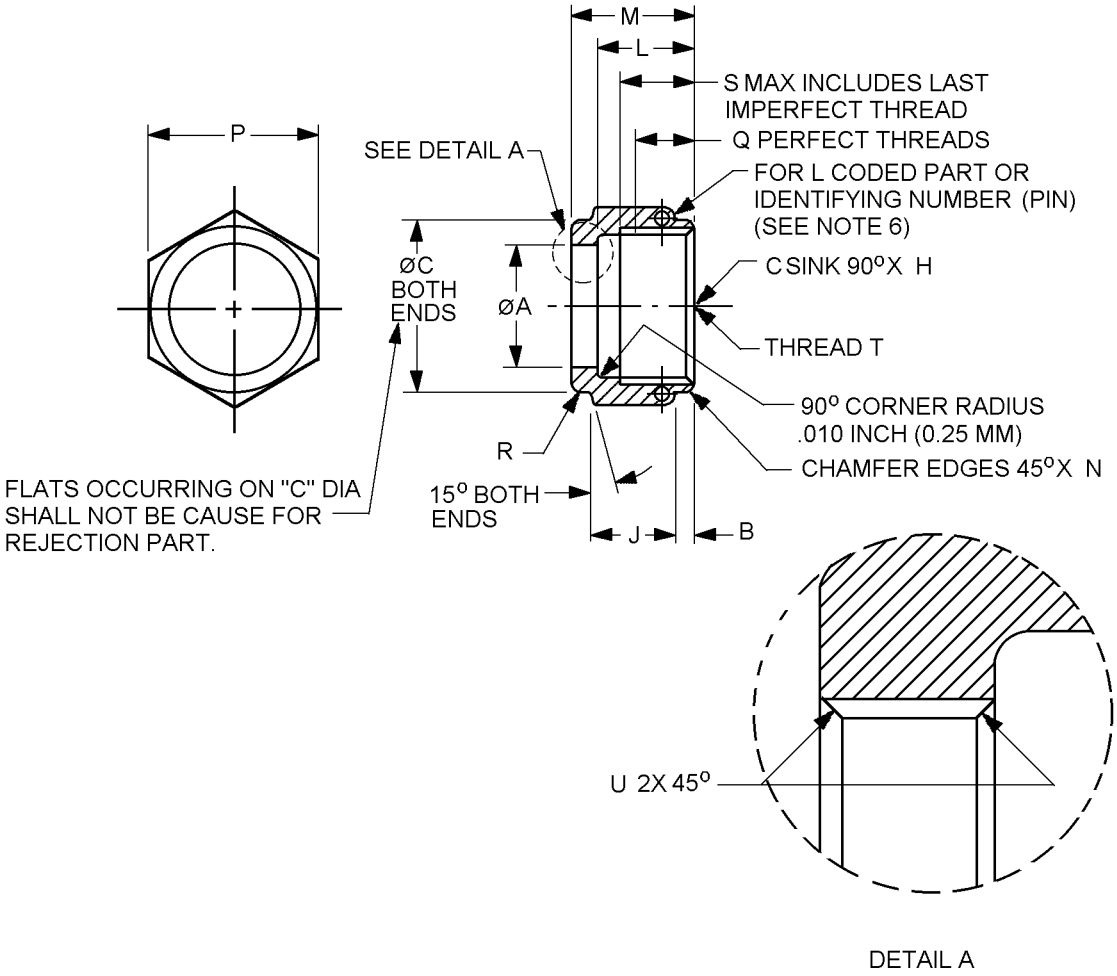


FIGURE 1. Nut, tube coupling.

AMSC N/A
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FSC 4730

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Dash number	Tube OD inches (mm)	Thread T SAE AS8879 (See note 5)	A dia. +.003 (0.08) -.000 inches (mm)	B inches (mm)	C dia. +.000 -.010 (0.3) inches (mm)
-2	.125 (3.18)	.3125 - 24UNJF -3B	.180 (4.57)	.109 (2.77)	.375 (9.53)
-3	.188 (4.78)	.3750 - 24UNJF -3B	.242 (6.15)	.141 (3.58)	.437 (11.10)
-4	.250 (6.35)	.4375 - 20UNJF -3B	.305 (7.75)	.125 (3.18)	.562 (14.27)
-5	.313 (7.95)	.5000 - 20UNJF -3B	.374 (9.50)	.125 (3.18)	.625 (15.88)
-6	.375 (9.53)	.5625 - 18UNJF - 3B	.440 (11.18)	.156 (3.96)	.687 (17.45)
-8	.500 (12.70)	.7500 - 16UNJF - 3B	.570 (14.48)	.156 (3.96)	.875 (22.23)
-10	.625 (15.88)	.8750 - 14UNJF - 3B	.698 (17.73)	.172 (4.37)	1.000 (25.40)
-12	.750 (19.05)	1.0625 - 12UNJ - 3B	.834 (21.18)	.125 (3.18)	1.250 (31.75)
-16	1.000 (25.40)	1.3125 - 12UNJ - 3B	1.089 (27.66)	.219 (5.56)	1.500 (38.10)
-20	1.250 (31.75)	1.6250 - 12UNJ - 3B	1.347 (34.21)	.312 (7.92)	2.000 (50.80)
-24	1.500 (38.10)	1.8750 - 12UNJ - 3B	1.617 (41.07)	.359 (9.12)	2.250 (57.15)
-28	1.750 (44.45)	2.2500 - 12UNJ - 3B	1.890 (48.01)	.375 (9.53)	2.625 (66.68)
-32	2.000 (50.80)	2.5000 - 12UNJ - 3B	2.167 (55.04)	.438 (11.13)	2.875 (73.03)
-40	2.500 (63.50)	3.0000 - 12UNJ - 3B	2.667 (67.74)	.406 (10.31)	3.375 (85.73)
-48	3.000 (76.20)	3.5000 - 12UNJ - 3B	3.180 (80.77)	.438 (11.13)	4.000 (101.60)

Dash number	H dia. +.031 (0.79) -.000 inches (mm)	J inches (mm)	L inches (mm)	M inches (mm)		N
-2	.312 (7.92)	.312 (7.92)	.460 (11.68)	.542 (13.77)	±.005 (0.13)	.010 (0.25)
-3	.375 (9.53)	.375 (9.53)	.506 (12.85)	.604 (15.34)		
-4	.438 (11.13)	.406 (10.31)	.532 (13.51)	.620 (15.75)		
-5	.500 (12.70)	.484 (12.39)	.579 (14.71)	.683 (17.35)		
-6	.562 (14.27)	.500 (12.70)	.603 (15.32)	.724 (18.39)	±.010 (0.25)	
-8	.750 (19.05)	.578 (14.68)	.723 (18.36)	.849 (21.56)		
-10	.875 (22.23)	.688 (17.48)	.817 (20.75)	.974 (24.74)		
-12	1.062 (26.97)	.750 (19.05)	.868 (22.05)	1.021 (25.93)		
-16	1.312 (33.32)	.766 (19.46)	.962 (24.43)	1.131 (28.73)		
-20	1.625 (41.28)	.766 (19.46)	1.017 (25.83)	1.224 (31.09)		
-24	1.875 (47.63)	.875 (22.23)	1.170 (29.72)	1.412 (35.86)		
-28	2.250 (57.15)	1.000 (25.40)	1.316 (33.43)	1.568 (39.83)		.015 (0.38)
-32	2.500 (63.50)	1.078 (27.38)	1.462 (37.13)	1.740 (44.20)		
-40	3.000 (76.20)	.969 (24.61)	1.328 (33.73)	1.631 (41.43)		
-48	3.500 (88.90)	1.063 (27.00)	1.452 (36.88)	1.756 (44.60)		

FIGURE 1. Nut, tube coupling - Continued.

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Dash number	P (Hex) inches (mm)		Q min inches (mm)	R rad inches (mm)	S max inches (mm)	U inches (mm)
-2	.375 (9.53)	+.003 (0.08) -.004 (0.10)	.251 (6.38)	.031 (0.79)	.376 (9.55)	.005 (0.13)
-3	.438 (11.13)		.338 (8.59)	.031 (0.79)	.428 (10.87)	
-4	.563 (14.30)		.380 (9.65)	.031 (0.79)	.500 (12.70)	
-5	.625 (15.88)		.401 (10.19)	.031 (0.79)	.542 (13.77)	
-6	.688 (17.48)		.413 (10.49)	.047 (1.19)	.569 (14.45)	.010 (0.25)
-8	.875 (22.23)		.484 (12.29)	.047 (1.19)	.647 (16.43)	
-10	1.000 (25.40)		.542 (13.77)	.047 (1.19)	.727 (18.47)	
-12	1.250 (31.75)	+.003 (0.08) -.006 (0.15)	.537 (13.64)	.062 (1.57)	.753 (19.13)	
-16	1.500 (38.10)		.563 (14.30)	.078 (1.98)	.813 (20.65)	
-20	2.000 (50.80)	±.016 (0.41)		.094 (2.39)		
-24	2.250 (57.15)	±.020 (0.51)	.661 (16.79)	.094 (2.39)	.911 (23.14)	
-28	2.625 (66.68)		.730 (18.54)	.094 (2.39)	.930 (23.62)	.015 (0.38)
-32	2.875 (73.03)		.844 (21.44)	.109 (2.77)	1.094 (27.79)	
-40	3.375 (85.73)		.718 (18.24)	.109 (2.77)	.968 (24.59)	
-48	4.000 (101.60)		.781 (19.84)	.109 (2.77)	1.031 (26.19)	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances: Decimals ±.005 inch (0.13 mm), angles ±5°.
4. Radius or break all corners .005 +.005/-.000 inch (0.13 +0.13/-0.00 mm).
5. Dimension A and thread T shall be concentric within .005 inch (0.13 mm) full indicator movement.
6. Remove all burrs and slivers.
7. Safety wire holes and location in accordance with SAE-AS1043.
8. For design features purposes, this standard takes precedence over documents referenced herein.

FIGURE 1. Nut, tube coupling - Continued.

REQUIREMENTS:

Dimensions and configurations: See figure 1.
Materials: See table I, unless otherwise specified finishes shall be alloys in accordance with SAE-AS4841.

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TABLE I. Material and nut alloys.

Nut AN818		
Designator	Material	Alloy
Blank	Steel	4130 or 4140
A 1/	CRES	A286 per SAE AMS5731, SAE AMS5732, SAE AMS5734, or SAE AMS5737
D 2/	Aluminum	2014-T6 or 2024-T6 or T851
J	CRES	304
K	CRES	316
M 3/, 4/, 5/	Monel	Nickel-copper alloy (UNS N04400)
N 4/	Nickel Alloy	625 per SAE AMS5666
R	CRES	321 per SAE AMS-QQ-S-763 or SAE AMS5645
T 6/	Titanium	6AI-4V
W	Aluminum	7075-T73
Z 7/	Steel	4130 or 4140

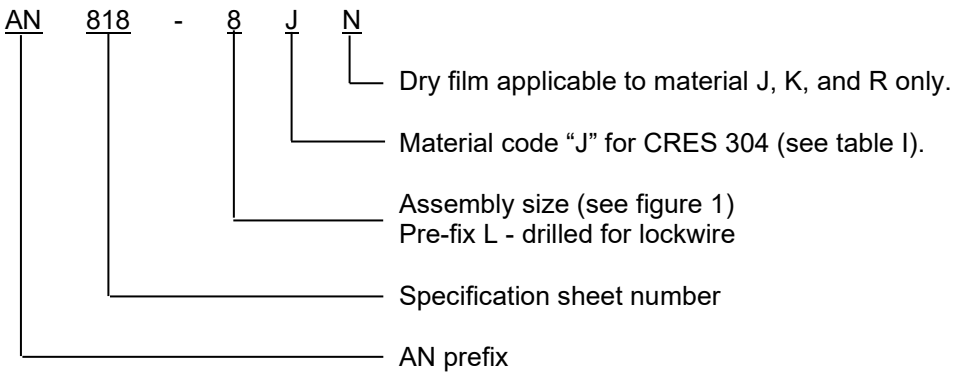
1/ Passivate per AMS2700, Type 2 or 8
2/ Inactive for new design.
3/ For use in Oxygen systems.
4/ No finish required.
5/ See ASTM B564 for forgings.
6/ Titanium shall not be used in oxygen systems.
7/ Zinc plate in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5.

Heat treatment. Steel alloys 4130 and 4140 only. Material in the solution treated condition, age to 28-40 HRC, 130 KSI UTS minimum.

Finish: Unless otherwise specified finishes shall be accordance with SAE-AS4841.

When specified dry film lubricant in accordance with MIL-PRF-46010 on all internal surfaces. Applicable only to material designators J, K, or R. Over spray on C dimension is permitted, see figure 1.

PIN:



PIN Examples:

AN818-8W identifies a nut assembly .500 inch (tube), without lock wire holes, aluminum (7075-T73).
AN818-L8K identifies a nut assembly .500 inch (tube), drilled for lockwire, CRES (316).

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AN818-L8JN identifies a nut assembly .500 inch (tube), with lock wire holes, CRES (304), coated internally with solid film lubricant (J, K, and R materials only).

Supersession data: Aluminum alloys 2014 and 2024 “D” designator is inactive for new design. For new design use aluminum alloy 7075 “W” designator to improve stress corrosion resistance and tensile strength.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to SAE-AS4841, this document references the following:

MIL-PRF-46010	SAE AMS 5732
ASTM B633	SAE AMS 5734
ASTM B695	SAE AMS 5737
ASTM B564	SAE AMS 5645
SAE AMS-QQ-S-763	SAE AMS 2700
SAE AMS 5666	SAE AS1043
SAE AMS 5731	SAE AS8879

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2024-009)

Review activities:
Army - CR4

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